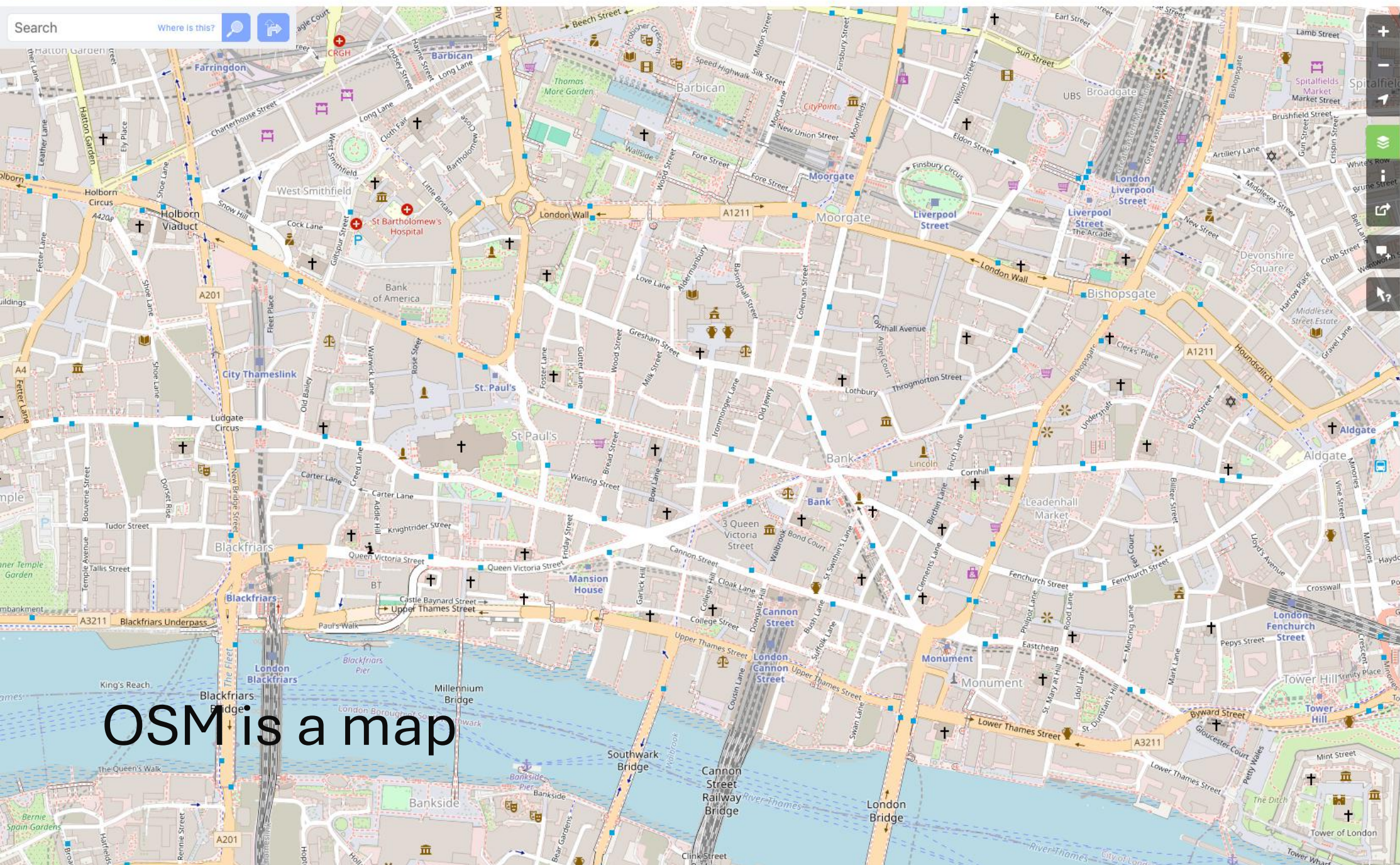




“The Wikipedia of the maps”

OpenStreetMap

<https://openstreetmap.org>



Map Layers

Standard

CyclOSM

Cycle Map

Transport Map

Tracestrack Topo

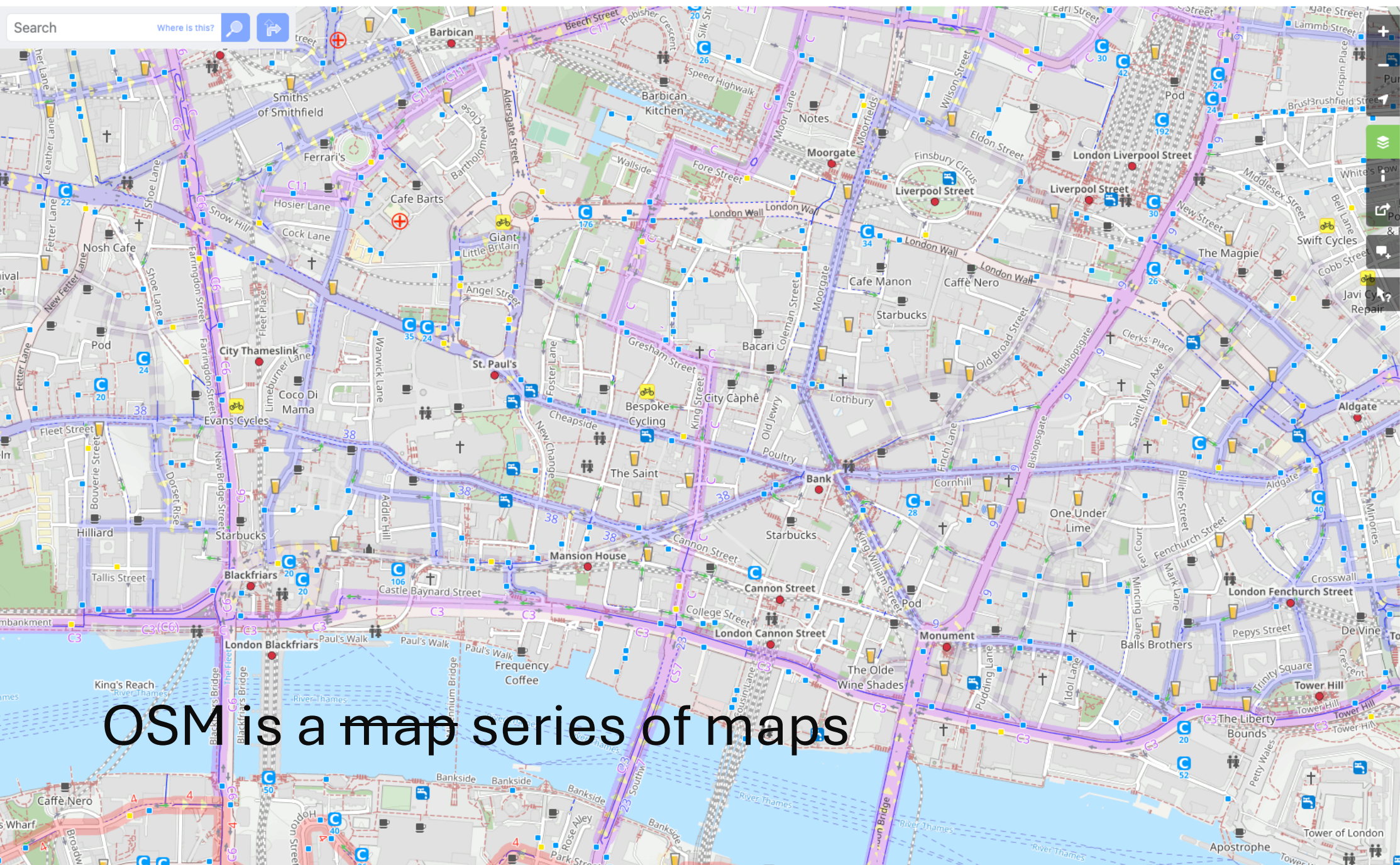
Humanitarian

Enable overlays for troubleshooting the map

☐ Map Notes

☐ Map Data

☐ Public GPS Traces



Map Layers

Standard

CyclOSM

Cycle Map

Transport Map

Tracestrack Topo

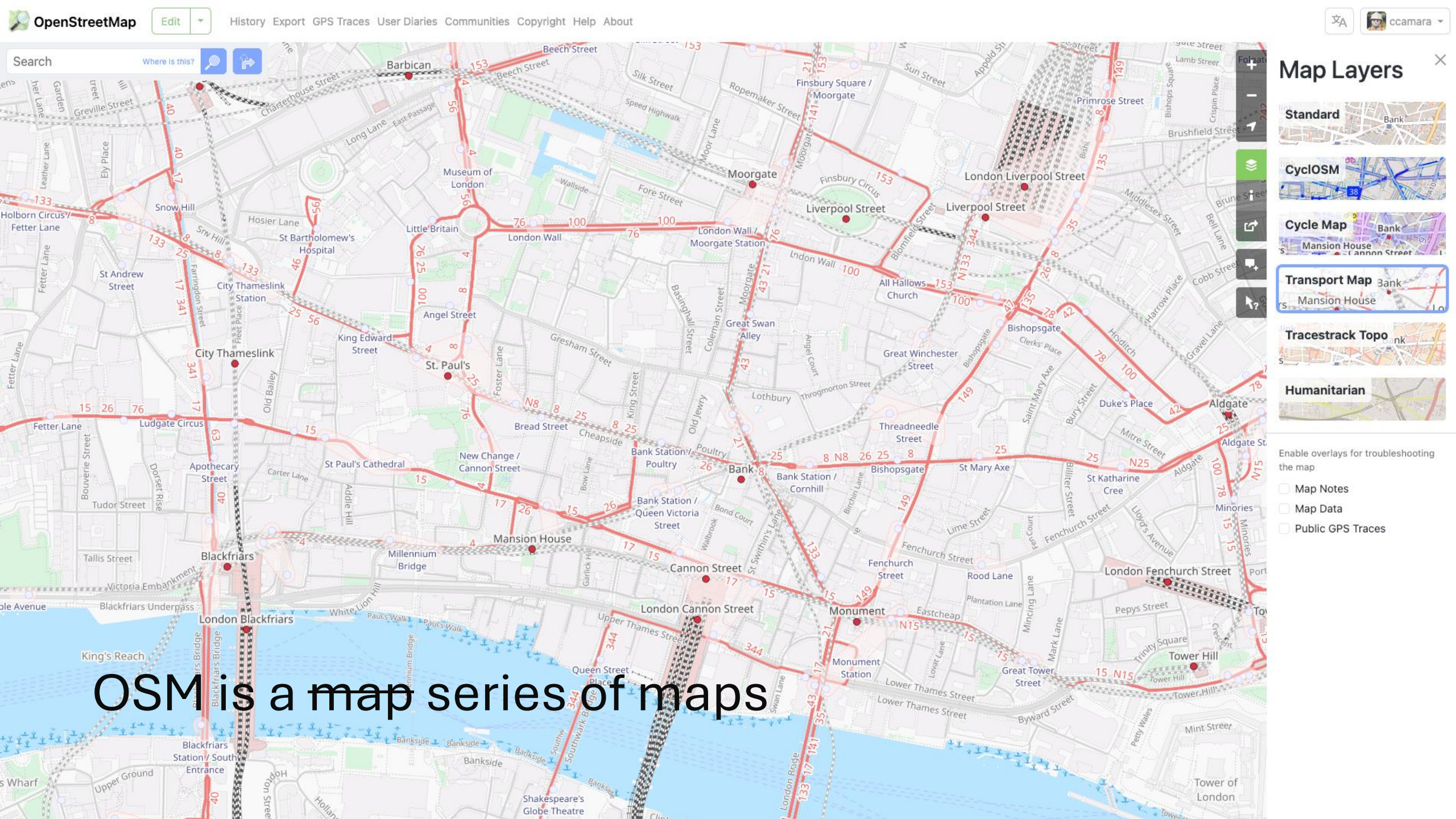
Humanitarian

Enable overlays for troubleshooting the map

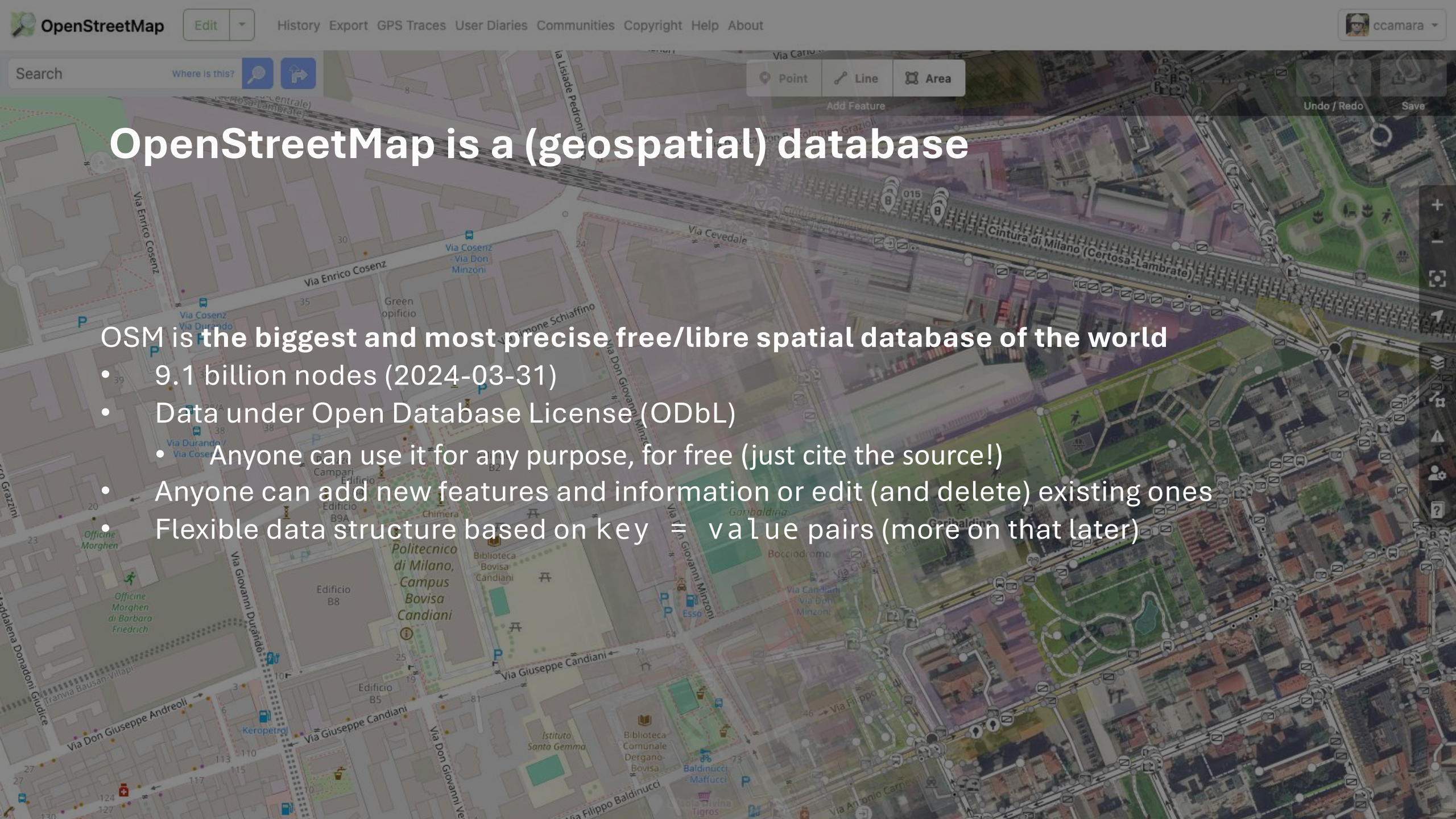
☐ Map Notes

☐ Map Data

☐ Public GPS Traces



OSM is a map series of maps



Search

Where is this?



Point

Line

Area

Add Feature

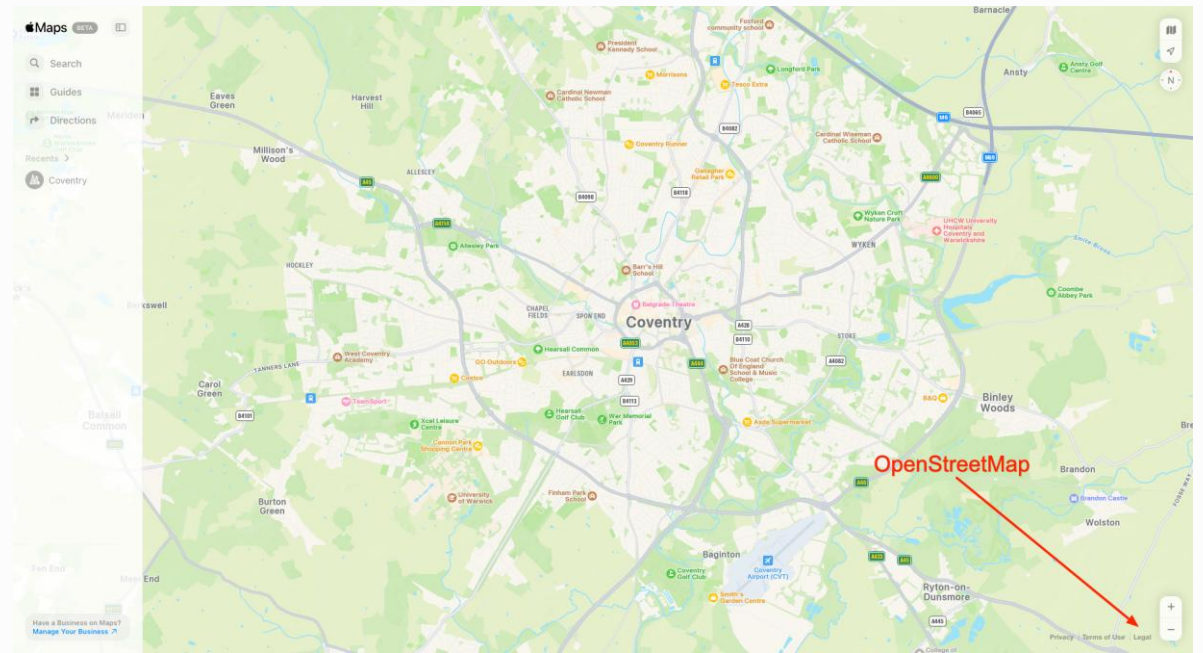
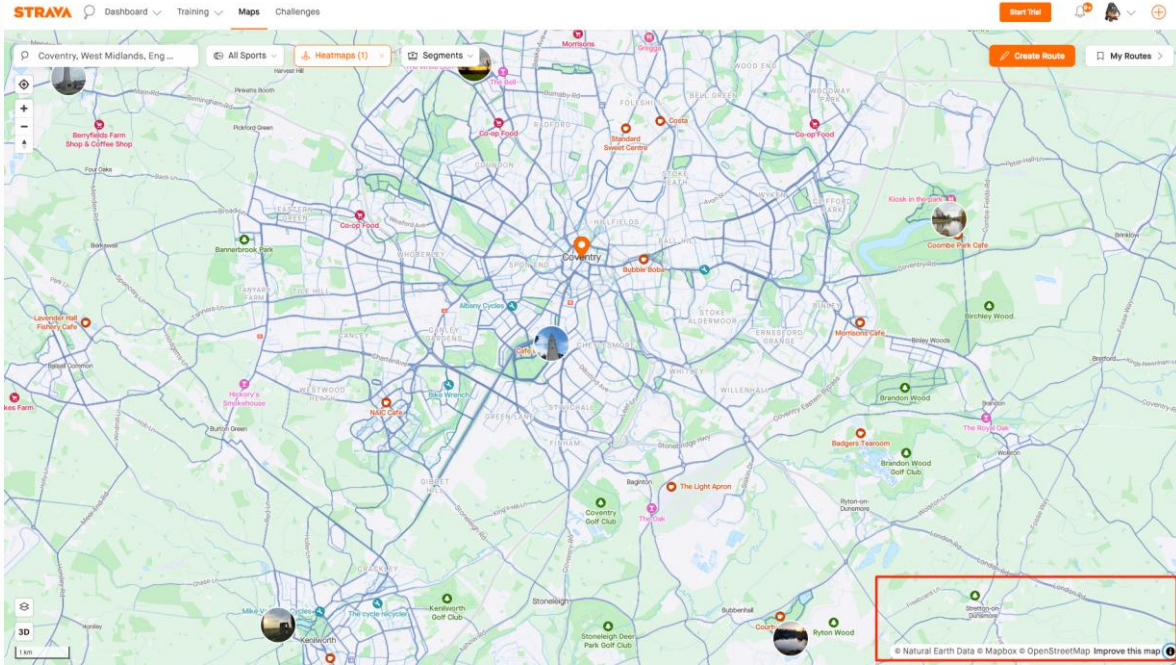
Undo / Redo

Save

OpenStreetMap is a (geospatial) database

OSM is the biggest and most precise free/libre spatial database of the world

- 9.1 billion nodes (2024-03-31)
- Data under Open Database License (ODbL)
 - Anyone can use it for any purpose, for free (just cite the source!)
- Anyone can add new features and information or edit (and delete) existing ones
- Flexible data structure based on key = value pairs (more on that later)



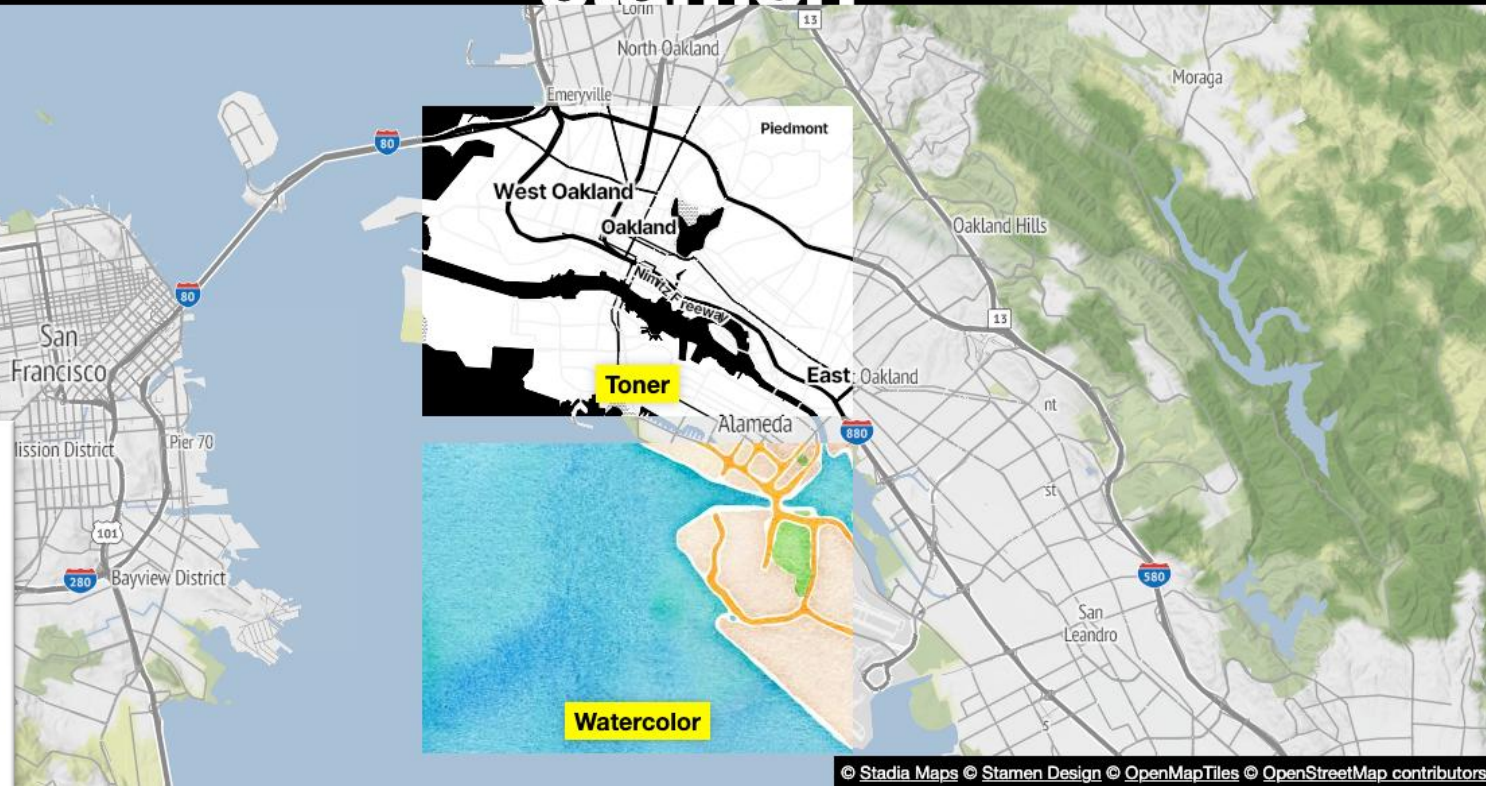
Its data, contributed by 10 million of volunteers, complements official data sources, and **populates thousands of tools and services**.

Major Sites: Amazon, Apple, Baidu Maps, Facebook, Microsoft, Wikipedia and Wikimedia

Transport: Air France, Alaska Airlines, Deutsche Bahn, Grab, SNCF (French rail agency), Uber

Geodata Software and Services: CARTO, Digital Globe, ESRI, Garmin, Mapbox, Telenav

Government: Agence Française de Développement, Government of Brazil, Government of Italy: President's Office, Police Scotland, US National Park Service, US State Department, USAID, Peace Corps....



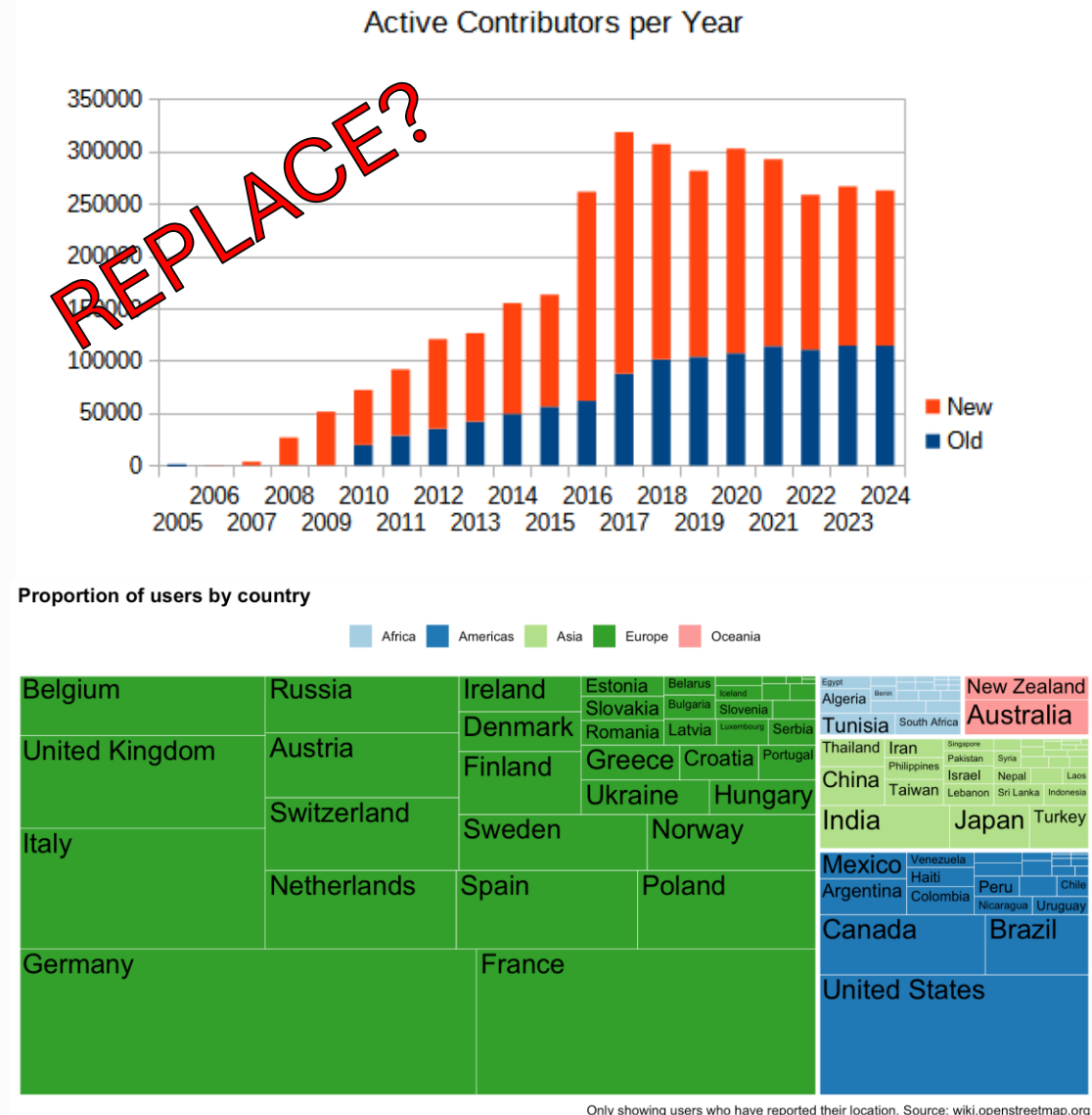
© Stadia Maps © Stamen Design © OpenMapTiles © OpenStreetMap contributors

Some of these companies create cartographic styles with OSM data for others to use: Stamen, Carto, Mapbox, Stadia, Mapzen...

OSM is a community (of communities)

More than **10 million users worldwide**, mainly volunteers (but not always)

- All users have **equal level of permissions** to:
 - Add new features to the map
 - Edit existing features to add more detail or remove outdated details
 - Remove
 - Decide what or how to organise the information in the dataset (i.e. what is included and what is not)
- Users may be part of local communities
- The OSM Foundation oversees the OSM project and has several WG



Like Wikipedia, **OSM is based on principles of openness and neutrality.**

“OpenStreetMap maps world as it exists, and includes mapping borders and countries according to actual current situation and **not a preferred or ideal situation**” ([OSM Wiki](#))

What can be mapped?



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Page

[Discussion](#)

Read

[Edit](#)

[Edit source](#)



More ▾



Scope

OpenStreetMap welcomes diverse contributions from our diverse community. Defining what falls within OSM remit is, therefore, challenging. As a rule of thumb, anything that ticks all the following conditions would be welcome:

- It is [verifiable](#),
- It is [mappable](#), within the planet [Earth](#),
- It is [not private](#) or copyrighted information,
- It is compatible with the [OpenStreetMap License](#),
- It is added manually (see the community guidelines for [automated edits](#)),
- It is not listed under [Limitations](#).

Limitations imposed by policies [[edit](#) | [edit source](#)]

License [[edit](#) | [edit source](#)]

All data in OpenStreetMap must be licensed under the [Open Database License](#) [↗] (ODbL).

OpenStreetMap is intentionally very strict about this by taking a [whiter than white](#) approach. For example [imports](#) are only allowed if the source has given OpenStreetMap explicit permission to use the data.

Whoever is using OSM data is obligated to follow its license, that may put some limitations - for example typical use of OSM data will require clearly crediting source visible to typical user.

Privacy [[edit](#) | [edit source](#)]

OpenStreetMap does not map where individual people live, how to contact individual people or who owns which building or plot. Private backyards are also not mapped in detail. For the full policy, please see [Mapping private information](#).

Not everything is mappable in OSM [[edit](#) | [edit source](#)]

While OpenStreetMap does not have any notability requirements, there are some limitations very much agreed upon:

- features that no longer exist should not be mapped in OpenStreetMap (but they may be mapped on the [Open Historical Map](#))
- temporary features are not mapped because our data is often downloaded and used offline for several weeks or months
 - certain events that happen in a regular pattern, like a weekly market, can be mapped by using different [time tags](#)
 - intermittent features, especially lakes and waterways are mappable with `intermittent=yes`
- [amenities](#) or rooms that are inaccessible to the public are not mapped (e.g. televisions, washing machines, showers or toilets in private apartments/houses).
 - See [Limitations on mapping private information](#)
- Certain useful but quickly changing data like live traffic info is out of scope of OSM

Verifiability [[edit](#) | [edit source](#)]

Main article: [Verifiability](#)

OSM data should, as far as is reasonably possible, be verifiable. The principle applies to tags and other aspects of data representation and essentially means another mapper should be able to come to the same place and collect the same data ("verify" the data you have entered). This principle excludes hypothetical or opinionated data like personal ratings and reviews.

For mapping fictional features one may use [Geofiction](#) - though they have own set of limitations and rules.

Any tags you like [[edit](#) | [edit source](#)]

OpenStreetMap lets you use [any tags you like](#). While this is very much the absence of a limitation it does introduce some possible difficulties in using OpenStreetMap data: it can be tricky to discover how to extract interesting data, and this can suddenly break because people started using a new tagging scheme and deprecated old one.

Tags can contain accidental typos. Tags do not have to be documented anywhere (although [documenting your custom tags is very much recommended](#)). New tags do not need to be discussed with anyone before being used (but you are very welcome to do so [in the forum](#) [↗], [the mailing list](#) [↗] or even by creating a [proposal](#)). Even [deprecated tags](#) can still be added, although you should have a good reason for doing so because tags are usually only deprecated if there is a very good reason for doing so.

Could these types of data be included in OSM?

- Population's data (e.g. census)
- Historical data
- Descriptions about a shop
 - menus,
 - Stairs
 - LGTBIQ+ friendliness
 - ratings
- Data from governments, city councils, police...

How are things mapped?

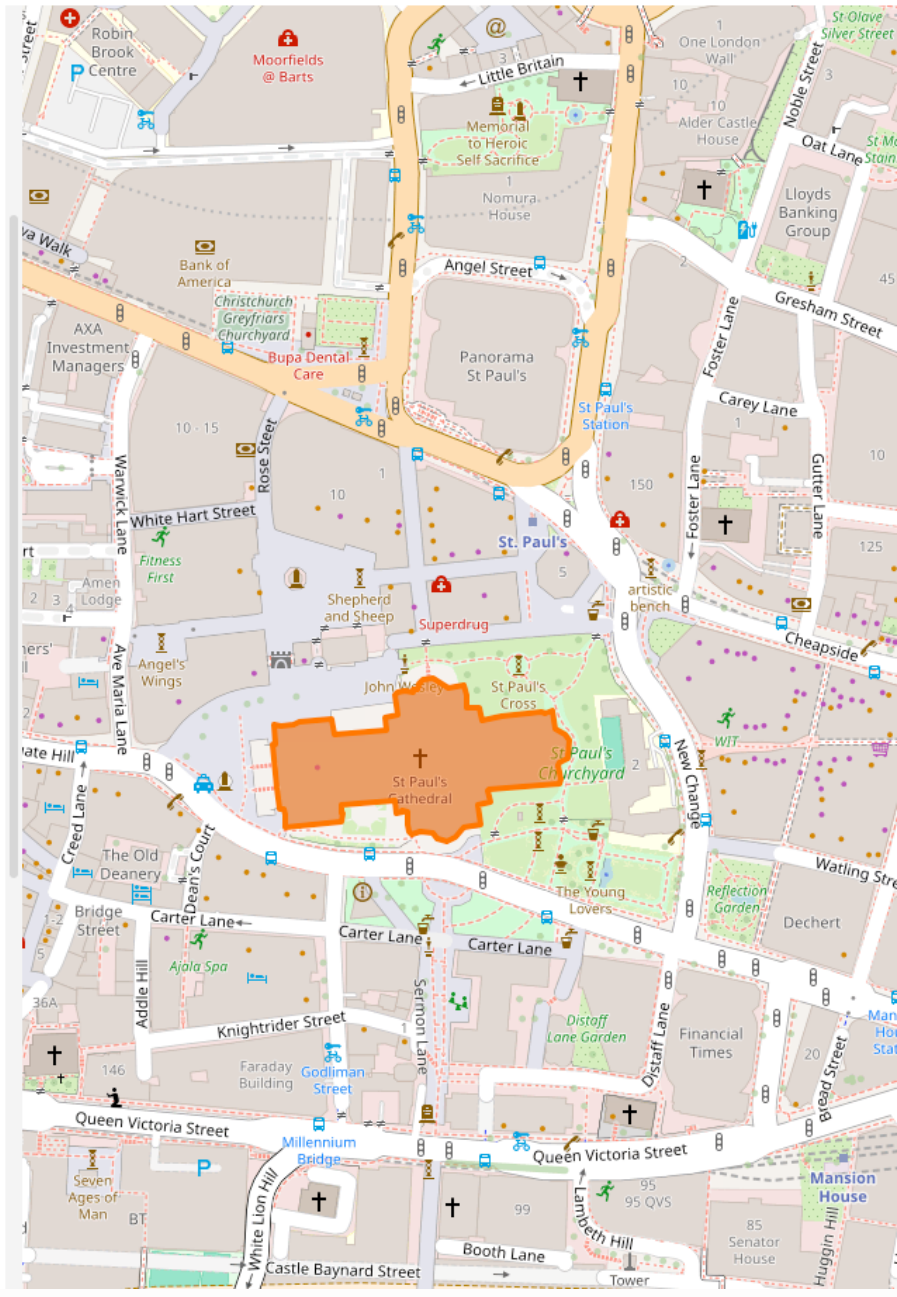
Understanding OSMs data structure

- In OSM, a **feature** is a physical element in the world subject to be mapped (street, building, sign, tree, bench...).
- Each OSM feature is described in two dimensions:
 - Spatial: through their coordinates (latitude and longitude) and element type (point, line, polygon, relationships)
 - **Meaning**: through tags (free format text fields)
- Each **tag** is defined by a pair of **key** = **value**
 - **Keys** can be assimilated to categories or topics
 - **Value** provide detail for the key-specified feature
- Not all keys are displayed on a map (rendered)
 - There's way more information that remains hidden (usually, for good reasons)
 - This information can be queried!

OpenStreetMap Edit History Export GPS Traces User Diaries Communities Copyright Help About

Tags

addr:city	London
addr:house:1	1
addr:house:2	2
addr:house:3	3
addr:house:4	4
addr:house:5	5
addr:house:6	6
addr:house:7	7
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addr:house:10	10
addr:house:11	11
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addr:house:94	94
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addr:house:96	96
addr:house:97	97
addr:house:98	98
addr:house:99	99
addr:house:100	100



key value

Querying OSM data

Using Overpass Turbo: <https://overpass-turbo.eu/>

▶ Run

↺

↻ Share

📄 Export

🔧 Wizard

🎨 Style

💾 Save

📂 Load

⚙ Settings

❓ Help

overpass turbo

Map

Data

```
1  /*
2  This has been generated by the overpass-turbo wizard.
3  The original search was:
4  "Drinking Water"
5  */
6  [out:json][timeout:25];
7  // gather results
8  node["amenity"="drinking_water"]({{bbox}});
9  // print results
10 out geom;
```

This data can be exported (downloaded) and used to create our own spatial dataset (foundation for a map)

How do we know what to query in Overpass Turbo?

In short: OSM Wiki and Taginfo

OSM wiki: collaborative documentation

<https://wiki.openstreetmap.org>

The wiki describes the consensus reached by OSM community.

Tags are described in OSM wiki. The wiki explains how to use them:

- Description/meaning
- Applicable elements
- List of accepted values
- Whether it is rendered or not
- ...

We can use the search feature to find how to tag certain features.



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Recent changes

Tools
What links here
Related changes
Upload file
Special pages
Printable version
Permanent link
Page information
Cite this page
Data item

Page [Discussion](#)

Read

[Edit](#)

[Edit source](#)

[View history](#)

[☆](#)

More ▾

[Search OpenStreetMap V](#) [Q](#)

Tag:amenity=drinking_water

(Redirected from [Drinking Water](#))

Tag:amenity=drinking water · In other languages

[Help](#)

[čeština](#) [Deutsch](#) **English** [español](#) [français](#) [Nederlands](#) [polski](#) [português](#) [Ελληνικά](#) [русский](#) [українська](#) [한국어](#) [中文（繁體）](#) [日本語](#)

[Other languages...](#)

Drinking water is a place where humans can obtain potable water for consumption. Typically, the water is used for drinking. This includes [drinking fountains or bubblers](#), water taps, water wells, springs and water points. Generally, this feature provides small quantities of water.

Where drinking water is provided in large quantities, use [amenity=water_point](#) or [waterway=water_point](#); i.e., for refilling tanks in motorhomes, boats, etc.

Contents [\[hide\]](#)

- 1 [How to map](#)
 - 1.1 [Useful combinations](#)
 - 1.2 [Nonworking features](#)
- 2 [Alternative tagging](#)
- 3 [Photo examples](#)
- 4 [Proposal history](#)
- 5 [External links](#)
- 6 [Common/possible tagging mistakes](#)
- 7 [See also](#)

How to map [\[edit\]](#) [\[edit source\]](#)

Add [amenity=drinking_water](#) to an existing element or create a new one.

 **amenity = drinking_water** [v](#) [d](#) [e](#)



Description

A drinking water source which provides potable water for consumption. [✎](#)

Rendering in **OSM Carto**



Group: [amenities](#)

Used on these [elements](#)



Implies

• [drinking_water=yes](#)

<https://taginfo.openstreetmap.org/>

Taginfo describes the actual uses of tags in the OSM (de facto).

- Most popular keys
- All possible values for a given key
- When possible, links to the wiki for more information

[illegible]



English

[KEY]=[VALUE]

[KEYS](#) · [TAGS](#) · [RELATIONS](#) · [PROJECTS](#) · [REPORTS](#) · [SOURCES](#) · [ABOUT](#)

building

To mark the outline of a building, a man-made structure with a roof, standing more or less permanently in one place.

XAPI

JOSM

Level0 Editor

Overpass turbo

OSM Tag History

ohsome

Overview

Values

Combinations

Similar

Chronology

Map

Wiki

Projects

Characters

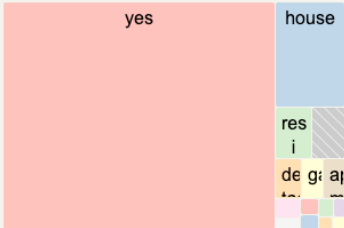
Overview

Statistics

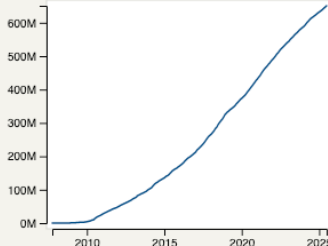
Type	Number of objects		Values
All	650 706 950	5.85%	7 866
Node	956 686	0.36%	1 069
Way	648 647 561	58.21%	7 101
Relation	1 102 703	8.20%	799

Objects with this key were last edited by 875 925 different users.

Distribution of values



Chronology



Status

Tag status from wiki pages: approved

Taginfo knows 53 projects referencing this key.

Wiki

→ Key:building

bg ca cs de en es fr hu it ja

ko nl no no pl pnb pt pt-br ru

uk yue zh-hans zh-hant

